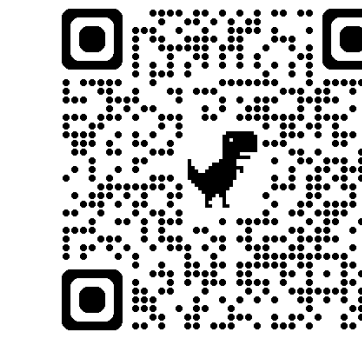




Motivation & Task Definition

Natural Interaction is Hard to Measure

Children's gaze behavior unfolds during free, face-to-face play and assessment. Child-worn sensors or fixed cameras can disrupt the interaction or miss important moments.



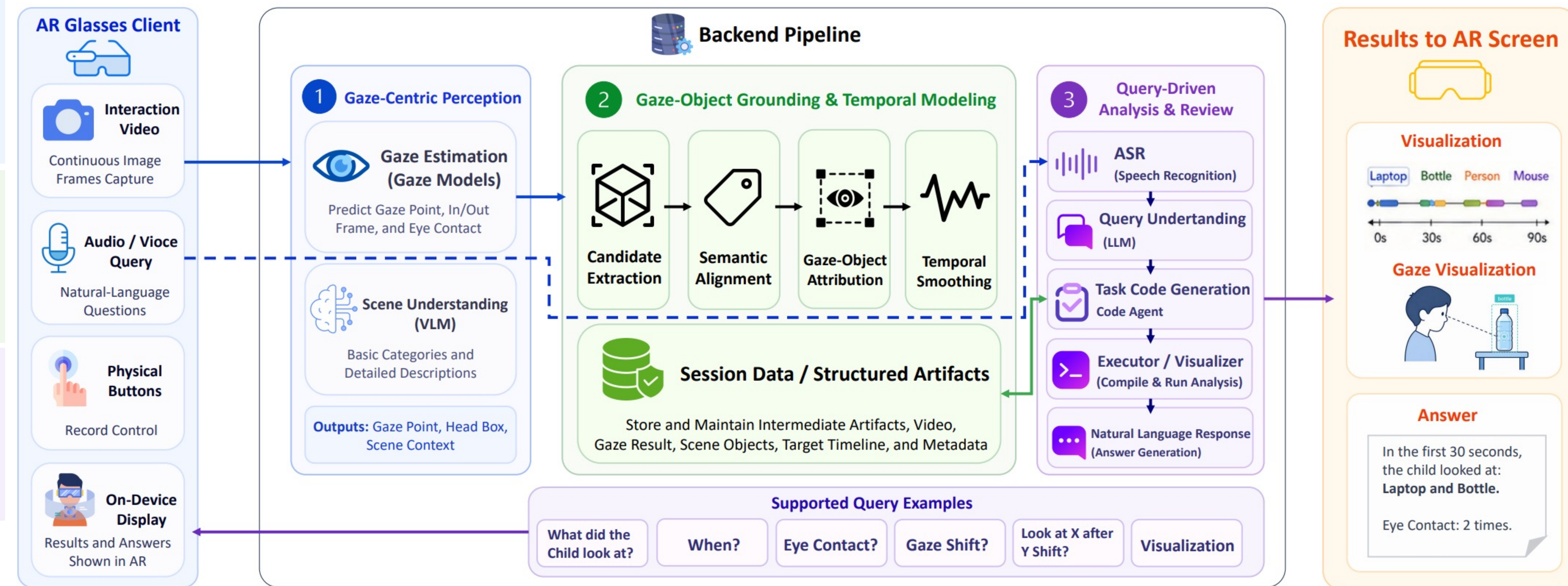
Manual Video Labeling is Slow

Current workflows often require practitioners to scrub through long videos or inspect timelines manually. This makes gaze analysis delayed, labor-intensive, and hard to scale.

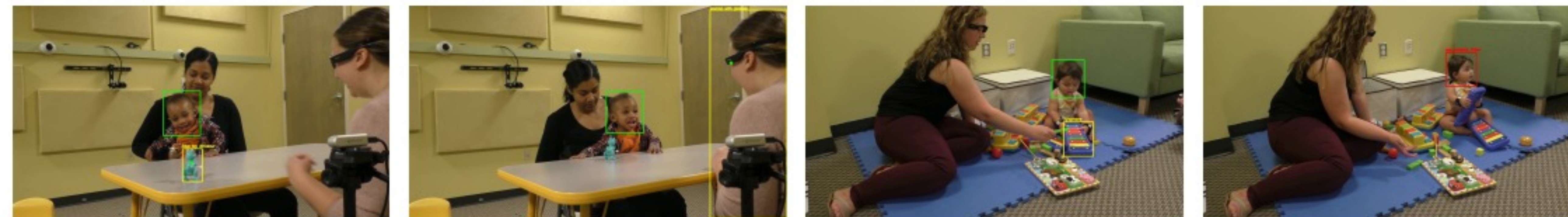
Raw Gaze Points from Gaze Model are not Enough

A gaze coordinate only says *where* the child looked, not *what it means*. Clinicians care whether the child looked at a toy, a face, or shifted attention between people and objects.

GazeAgent Architecture



Example Queries & Output



Q: When did the child look at the red toy car?

A: The child looked at the red toy car **3 times**: 00:12–00:30, 00:35–01:00, 01:18–01:30

Q: How many times did the child had eye contact with me?

A: The child perform 2 eye contacts

Q: Show me all of the child's gaze shifts

A: The child's gaze shifted across eye contact, red toy car, green dinosaur, and blocks.

Blocks → Eye Contact at 00:10; Eye contact → Red toy car at 00:12; Red toy car → Eye contact at 00:30; Eye contact → Red toy car at 00:35; Red toy car → Green dinosaur toy at 01:00

1 Capture

Interaction Capture of Natural Interaction



Continuous recording of what the Clinician sees and hears

2 Query

Ask Contextual Questions About Gaze Behavior

Did the child look at the red toy car?
 When did the child look at me?
 How did the child's attention shift during this interaction?
 ...

GazeAgent

Understand → Ground → Reason
 (Scene understanding, gaze grounding, temporal reasoning)

3 Insight

Receive Concise, Interpretable Summaries

Looked at red toy car
 3 times Total duration: 55s
 00:12-00:30 00:35-01:00 01:18-01:30

Looked at clinician (eye contact)
 2 events Total duration: 7s
 00:10-00:12 00:30-00:35

